EngageIQ — Smart Engagement Opportunity Scorer

BAX-423 Big Data · Spring 2026 · Yang, Alice · UC Davis GSM

Problem Statement

Developers and founders waste hours manually scanning GitHub and Hacker News to find where to contribute, comment, or build reputation. EngageIQ automates this by ingesting 10,046 opportunities across 15 technical domains, scoring them via semantic embeddings and community signals, and adapting rankings in real time from user feedback.

System Architecture

Stage	Component	Lecture
1 · Ingestion	GitHub API + Algolia/HN API. Bloom Filter dedup. Python queue-based on-demand refresh. Real HN stories fetched by domain keyword queries (url_type: hn_item, 0 fallbacks).	Lec 2
2 · Embedding	Sentence-BERT all-MiniLM-L6-v2 (384-dim). FAISS IndexFlatIP exact search < 2 ms.	Lec 5
3 · Intent	7-intent classifier from role + interests. Adaptive query expansion + candidate injection.	app.py
4 · Scoring	Composite: 0.40 relevance + 0.30 community + 0.20 visibility + 0.10 (1-effort).	Lec 7
5 · Ranking	Multi-stage: intent-aware rerank + per-domain diversity cap + Thompson Sampling rerank.	Lec 7/8
6 · Adaptation	Thompson Sampling Beta-Bernoulli bandit. Domain preference learning over 50+ rounds.	Lec 8
7 · Analytics	Pandas batch: domain health, trending repos, volume-over-time, rising opportunities.	Lec 3
8 · Dashboard	Streamlit: ranked cards, Why this?, suggested actions, CSV/JSON export.	—

Dataset

Metric	Value
Total records	10,046
Technical domains	15
GitHub records	8,588 — real API-derived issues + repos; direct github.com links
HN records	1,458 — real Algolia/HN API stories; direct news.ycombinator.com/item?id= URLs; 0 fallbacks
Storage	CSV offline snapshot (committed to repo) + SQLite for live records
Live / offline split	10,046 offline snapshot; live records added via on-demand API refresh

BAX-423 Techniques & Benchmarks

Integrated BAX-423 Techniques

Technique	Lec	File	Role in System	Benchmark
Bloom Filter (sketching/dedup)	2	bloom_filter.py	Rejects duplicate streamed records before DB insert.	~0 collision rate on 10,046 unique IDs
Sentence-BERT + FAISS IndexFlatIP	5	embeddings.py	384-dim semantic embeddings. Exact inner-product search at 10k scale.	Query latency: <2 ms NDCG@10 (emb only): 0.49
Multi-stage Ranking (NDCG)	7	ranking.py	Composite score + diversity rerank. NDCG@10 outperforms stars-only by 55%.	NDCG@10: 0.2425 vs stars-only: 0.1569
Thompson Sampling (RL bandit)	8	adaptive_learning.py	Beta-Bernoulli posterior per opportunity. Domain preference learns from feedback.	50 rounds: ML/AI pref 0.50 -> 1.0

Ranking Benchmark — NDCG@10 (Sofia Persona, n=500, seed=42)

Graded relevance: in-domain records scored 0–1 by stars (40%) + activity (30%) + community health (20%) + GFI bonus (20%). Higher = better (max 1.0). Random stratified sample, seed=42.

Ranking Method	NDCG@10	Notes
Embedding Similarity Only	0.4915	Strong domain retrieval for aligned queries
Full Composite + Re-rank *	0.2425	**Composite scoring + diversity; beats stars-only by 55%**
Stars-Only Ranking	0.1569	Popularity does not equal persona relevance
Random Baseline	0.1036	No signal — lower bound

* Benchmark isolates composite scoring. In the live app, Full Composite also applies intent-specific GFI/diversity reranking on top of embedding retrieval — complementary stages.

Adaptive Learning — Thompson Sampling (50 Rounds)

Synthetic Sofia persona: engage if domain in {ML, AI Research}, skip otherwise. Beta posterior updated each round. Seed=42.

Metric	Before (Round 0)	After (Round 50)
ML + AI Research domain pref score	0.50 (uniform prior)	1.00 (maximum)
Other 13 domains avg pref score	0.50 (uniform prior)	0.00 (minimized)
Preferred / non-preferred score gap	0.00	+1.00

6 Core Capabilities & Persona Test Results

6 Core Capabilities

#	Capability	Implementation
1	**Multi-Source Ingestion & On-Demand Refresh**	GitHub API + Hacker News API. Bloom Filter dedup. Python queue-based live refresh.
2	**Content Embedding & Similarity Retrieval**	all-MiniLM-L6-v2 (384-dim). FAISS IndexFlatIP. Embedding cache for <2 ms queries.
3	**Engagement Scoring & Multi-Stage Ranking**	4-component composite + diversity rerank. NDCG@10 = 0.2425 vs 0.1036 random baseline.
4	**Adaptive Learning from Feedback**	Thompson Sampling Beta-Bernoulli bandit. Domain pref 0.50 -> 1.00 over 50 rounds.
5	**Batch Analytics & Trend Detection**	Pandas batch: domain health, trending repos, volume-over-time, rising opportunities.
6	**Dashboard & Engagement Brief**	Streamlit: ranked cards, Why this?, suggested actions. CSV/JSON export.

Persona Pass/Fail Test Results

Each persona tested against all 6 capabilities. Pass = feature surfaces expected results for that persona's interests.

Persona / Role	1	2	3	4	5	6	Result
Sofia — ML Student	Pass	Pass	Pass	Pass	Pass	Pass	**PASS**
David — DevOps Engineer	Pass	Pass	Pass	Pass	Pass	Pass	**PASS**
Lina — Data Journalist	Pass	Pass	Pass	Pass	Pass	Pass	**PASS**
Raj — Startup Founder	Pass	Pass	Pass	Pass	Pass	Pass	**PASS**

Evidence: Sofia top-10 surfaces ML/AI Research repos (gfi=6, domain_match=8). David leads with DevOps/K8s + Cloud API repos (infra_kw=4, domain_match=9). Lina includes trending open-source sorted by growth_rate (avg_trend=0.69, hn=10). Raj highlights Developer Tools and B2B SaaS (api=6, domain_match=10).

Hidden Persona Robustness — Adaptive Intent Layer (11/11 PASS)

The same intent inference, adaptive query expansion, and intent-aware reranking layer powers both the main Action Queue and the Persona Test Panel. Stress-tested against 11 roles not in ROLE_INTENT map — intent inferred from role + interest keywords. Pass criteria: domain_match >= 4, primary_match >= 1, src_fit >= 6, neg = 0.

Hidden Persona	Inferred Intent	domain	primary	src_fit	neg	Result
Security Researcher	security_review	5/10	5/10	10/10	0	**PASS**
Climate Tech Founder	startup_growth	7/10	4/10	10/10	0	**PASS**
Beginner Developer	contribution	5/10	4/10	10/10	0	**PASS**
Open Source Maintainer	community_engagement	8/10	4/10	10/10	0	**PASS**
Product Manager	startup_growth	7/10	7/10	10/10	0	**PASS**
Mobile Developer	mobile_contribution	8/10	8/10	10/10	0	**PASS**
Game Developer	generic	10/10	6/10	10/10	0	**PASS**

Data Engineer	data_engineering	6/10	5/10	9/10	0	**PASS**
Academic ML Researcher	generic	10/10	8/10	10/10	0	**PASS**
Education Creator	trend_spotting	10/10	9/10	10/10	0	**PASS**
Privacy Researcher	security_review	5/10	5/10	10/10	0	**PASS**

Limitations, Future Work & Submission Checklist

Honest Scope — Course-Project Implementation

Area	Current Implementation	Production Equivalent
Streaming	On-demand API refresh + Python queue/thread simulation.	Apache Kafka / Flink persistent streaming cluster.
Data Sources	GitHub + Hacker News. Reddit excluded — OAuth2 unavailable in scope.	Expand to Reddit PRAW, LinkedIn, DEV.to.
AI Suggestions	Deterministic template-based engagement action generator.	LLM-generated actions via Claude / GPT with prompt caching.
Batch Analytics	Pandas batch over 10,046-record offline snapshot (<1 s latency).	Apache Spark / Dask for distributed processing at scale.
GH Archive	build_real_dataset.py includes GH Archive support; not used in runtime.	Scheduled GH Archive pulls as supplemental source.

Future Work

• Connect suggested engagement actions to a live LLM (e.g., Claude Haiku) for natural-language drafts; cache generated briefs to minimize API cost.

• Expand data sources to Reddit (authenticated PRAW), LinkedIn, and DEV.to for broader community coverage.

• Add scheduled GH Archive pulls for historical trend analysis beyond real-time API limits.

• Extend Thompson Sampling to opportunity-type preferences (issue vs. repo vs. post) for finer-grained adaptive ranking.

• Replace Pandas batch analytics with Dask or Spark when dataset grows beyond 100k records.

Deployment Notes

App requires sentence-transformers and faiss-cpu (~450 MB combined). On first launch, embeddings are computed once and cached to data/embeddings.npy (~16 MB). Subsequent launches load the cache in < 1 second. Live API fetch is optional — the app runs fully on the pre-seeded offline dataset.

Submission Checklist

Item	Status
code/ — all .py files + requirements.txt	**Included**
data/opportunities.csv — 10,046 offline records	**Included**
data/embeddings.npy — pre-computed 384-dim embeddings	**Included**
brief.pdf — this document	**Included**
prompts.md — development + planned runtime prompts	**Included**
Live public URL	https://engageiq-bax423git-qianyingyang.streamlit.app
GitHub Repo	https://github.com/qyayang/engageiq-bax423
ZIP: Yang_Alice_BAX423_Final.zip	**Per one-pager spec**